

Density and Specific Gravity

2-1C What is the difference between intensive and extensive properties?

2-2C What is specific gravity? How is it related to density?

2-3C Under what conditions is the ideal-gas assumption suitable for real gases?

2-4C What is the difference between R and R_u ? How are these two related?

2-5 A spherical balloon with a diameter of 6 m is filled with helium at 20°C and 200 kPa. Determine the mole number and the mass of the helium in the balloon. *Answers: 9.28 kmol, 37.2 kg*

2-6  Reconsider Prob. 2-5. Using EES (or other) software, investigate the effect of the balloon diameter on the mass of helium contained in the balloon for the pressures of (a) 100 kPa and (b) 200 kPa. Let the diameter vary from 5 m to 15 m. Plot the mass of helium against the diameter for both cases.

2-7 The pressure in an automobile tire depends on the temperature of the air in the tire. When the air temperature is 25°C, the pressure gage reads 210 kPa. If the volume of the tire is 0.025 m³, determine the pressure rise in the tire when the air temperature in the tire rises to 50°C. Also, determine the amount of air that must be bled off to restore pressure to its original value at this temperature. Assume the atmospheric pressure to be 100 kPa.



FIGURE P2-7

* Problems designated by a “C” are concept questions, and students are encouraged to answer them all. Problems designated by an “E” are in English units, and the SI users can ignore them. Problems with the  icon are solved using EES, and complete solutions together with parametric studies are included on the enclosed DVD. Problems with the  icon are comprehensive in nature and are intended to be solved with a computer, preferably using the EES software that accompanies this text.

2-8E The air in an automobile tire with a volume of 0.53 ft³ is at 90°F and 20 psig. Determine the amount of air that must be added to raise the pressure to the recommended value of 30 psig. Assume the atmospheric pressure to be 14.6 psia and the temperature and the volume to remain constant.

Answer: 0.0260 lbm

2-9E A rigid tank contains 20 lbm of air at 20 psia and 70°F. More air is added to the tank until the pressure and temperature rise to 35 psia and 90°F, respectively. Determine the amount of air added to the tank. *Answer: 13.7 lbm*

2-10  The density of atmospheric air varies with elevation, decreasing with increasing altitude. (a) Using the data given in the table, obtain a relation for the variation of density with elevation, and calculate the density at an elevation of 7000 m. (b) Calculate the mass of the atmosphere using the correlation you obtained. Assume the earth to be a perfect sphere with a radius of 6377 km, and take the thickness of the atmosphere to be 25 km.

z , km	ρ , kg/m ³
6377	1.225
6378	1.112
6379	1.007
6380	0.9093
6381	0.8194
6382	0.7364
6383	0.6601
6385	0.5258
6387	0.4135
6392	0.1948
6397	0.08891
6402	0.04008

Vapor Pressure and Cavitation

2-11C What is vapor pressure? How is it related to saturation pressure?

2-12C Does water boil at higher temperatures at higher pressures? Explain.

2-13C If the pressure of a substance is increased during a boiling process, will the temperature also increase or will it remain constant? Why?

2-14C What is cavitation? What causes it?

2-15 In a piping system, the water temperature remains under 40°C. Determine the minimum pressure allowed in the system to avoid cavitation.

2-16 The analysis of a propeller that operates in water at 20°C shows that the pressure at the tips of the propeller drops to 2 kPa at high speeds. Determine if there is a danger of cavitation for this propeller.

2-17E The analysis of a propeller that operates in water at 70°F shows that the pressure at the tips of the propeller drops to 0.1 psia at high speeds. Determine if there is a danger of cavitation for this propeller.

2-18 A pump is used to transport water to a higher reservoir. If the water temperature is 25°C, determine the lowest pressure that can exist in the pump without cavitation.

Energy and Specific Heats

2-19C What is the difference between the macroscopic and microscopic forms of energy?

2-20C What is total energy? Identify the different forms of energy that constitute the total energy.

2-21C List the forms of energy that contribute to the internal energy of a system.

2-22C How are heat, internal energy, and thermal energy related to each other?

2-23C What is flow energy? Do fluids at rest possess any flow energy?

2-24C How do the energies of a flowing fluid and a fluid at rest compare? Name the specific forms of energy associated with each case.

2-25C Using average specific heats, explain how internal energy changes of ideal gases and incompressible substances can be determined.

2-26C Using average specific heats, explain how enthalpy changes of ideal gases and incompressible substances can be determined.

Coefficient of Compressibility

2-27C What does the coefficient of compressibility of a fluid represent? How does it differ from isothermal compressibility?

2-28C What does the coefficient of volume expansion of a fluid represent? How does it differ from the coefficient of compressibility?

2-29C Can the coefficient of compressibility of a fluid be negative? How about the coefficient of volume expansion?

2-30 It is observed that the density of an ideal gas decreases by 10 percent when compressed isothermally from 10 atm to 11 atm. Determine the percent decrease in density of the gas if it is compressed isothermally from 100 atm to 101 atm.

2-31 Using the definition of the coefficient of volume expansion and the expression $\beta_{\text{ideal gas}} = 1/T$, show that the percent increase in the specific volume of an ideal gas during isobaric expansion is equal to the percent increase in absolute temperature.

2-32 Water at 1 atm pressure is compressed to 800 atm pressure isothermally. Determine the increase in the density of water. Take the isothermal compressibility of water to be $4.80 \times 10^{-5} \text{ atm}^{-1}$.

2-33 Water at 15°C and 1 atm pressure is heated to 100°C at constant pressure. Using coefficient of volume expansion data, determine the change in the density of water.

Answer: -38.7 kg/m^3

2-34 Saturated refrigerant-134a liquid at 10°C is cooled to 0°C at constant pressure. Using coefficient of volume expansion data, determine the change in the density of the refrigerant.

2-35 A water tank is completely filled with liquid water at 20°C. The tank material is such that it can withstand tension caused by a volume expansion of 2 percent. Determine the maximum temperature rise allowed without jeopardizing safety.

2-36 Repeat Prob. 2-35 for a volume expansion of 1 percent for water.

2-37 The density of seawater at a free surface where the pressure is 98 kPa is approximately 1030 kg/m^3 . Taking the bulk modulus of elasticity of seawater to be $2.34 \times 10^9 \text{ N/m}^2$ and expressing variation of pressure with depth z as $dP = \rho g dz$ determine the density and pressure at a depth of 2500 m. Disregard the effect of temperature.

Viscosity

2-38C What is viscosity? What is the cause of it in liquids and in gases? Do liquids or gases have higher dynamic viscosities?

2-39C What is a Newtonian fluid? Is water a Newtonian fluid?

2-40C Consider two identical small glass balls dropped into two identical containers, one filled with water and the other with oil. Which ball will reach the bottom of the container first? Why?

2-41C How does the dynamic viscosity of (a) liquids and (b) gases vary with temperature?

2-42C How does the kinematic viscosity of (a) liquids and (b) gases vary with temperature?

2-43 A 50-cm $\times$ 30-cm $\times$ 20-cm block weighing 150 N is to be moved at a constant velocity of 0.8 m/s on an inclined surface with a friction coefficient of 0.27. (a) Determine the force F that needs to be applied in the horizontal direction. (b) If a 0.4-mm-thick oil film with a dynamic viscosity of $0.012 \text{ Pa} \cdot \text{s}$ is applied between the block and inclined surface, determine the percent reduction in the required force.

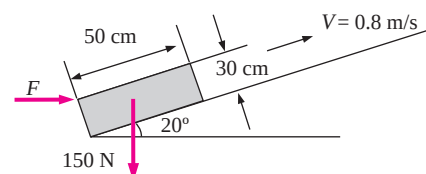


FIGURE P2-43

2-44 Consider the flow of a fluid with viscosity μ through a circular pipe. The velocity profile in the pipe is given as $u(r) = u_{\max}(1 - r^n/R^n)$, where $u_{\max}$ is the maximum flow velocity, which occurs at the centerline; r is the radial distance from the centerline; and $u(r)$ is the flow velocity at any position r . Develop a relation for the drag force exerted on the pipe wall by the fluid in the flow direction per unit length of the pipe.

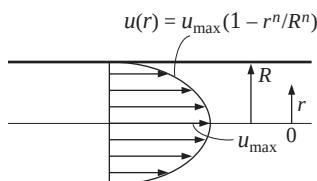


FIGURE P2-44

2-45 A thin 20-cm $\times$ 20-cm flat plate is pulled at 1 m/s horizontally through a 3.6-mm-thick oil layer sandwiched between two plates, one stationary and the other moving at a constant velocity of 0.3 m/s, as shown in Fig. P2-45. The dynamic viscosity of oil is 0.027 Pa $\cdot$ s. Assuming the velocity in each oil layer to vary linearly, (a) plot the velocity profile and find the location where the oil velocity is zero and (b) determine the force that needs to be applied on the plate to maintain this motion.

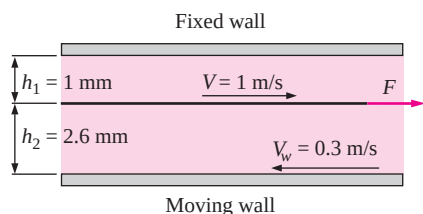


FIGURE P2-45

2-46 A frustum-shaped body is rotating at a constant angular speed of 200 rad/s in a container filled with SAE 10W oil

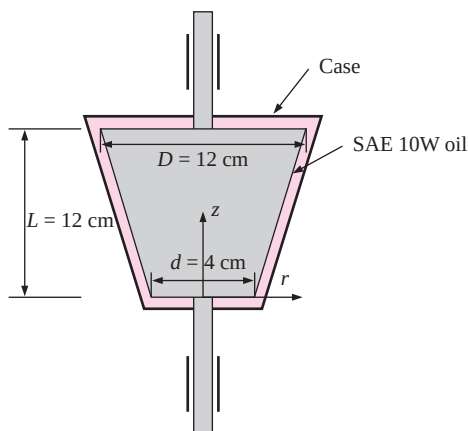


FIGURE P2-46

at 20°C ($\mu = 0.1$ Pa $\cdot$ s), as shown in Fig. P2-46. If the thickness of the oil film on all sides is 1.2 mm, determine the power required to maintain this motion. Also determine the reduction in the required power input when the oil temperature rises to 80°C ($\mu = 0.0078$ Pa $\cdot$ s).

2-47 The clutch system shown in Fig. P2-47 is used to transmit torque through a 3-mm-thick oil film with $\mu = 0.38$ N $\cdot$ s/m² between two identical 30-cm-diameter disks. When the driving shaft rotates at a speed of 1450 rpm, the driven shaft is observed to rotate at 1398 rpm. Assuming a linear velocity profile for the oil film, determine the transmitted torque.

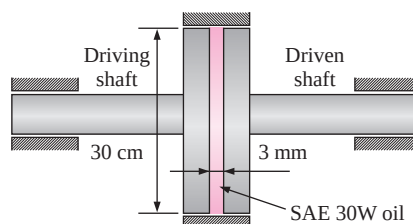


FIGURE P2-47

2-48  Reconsider Prob. 2-47. Using EES (or other) software, investigate the effect of oil film thickness on the torque transmitted. Let the film thickness vary from 0.1 mm to 10 mm. Plot your results, and state your conclusions.

2-49 The viscosity of some fluids changes when a strong electric field is applied on them. This phenomenon is known as the electrorheological (ER) effect, and fluids that exhibit such behavior are known as ER fluids. The Bingham plastic model for shear stress, which is expressed as $\tau = \tau_y + \mu(du/dy)$ is widely used to describe ER fluid behavior because of its simplicity. One of the most promising applications of ER fluids is the ER clutch. A typical multidisk ER clutch consists of several equally spaced steel disks of inner radius R_1 and outer radius R_2 , N of them attached to the input shaft. The gap h between the parallel disks is filled with a viscous fluid. (a) Find a relationship for the torque generated by the clutch when the output shaft is stationary

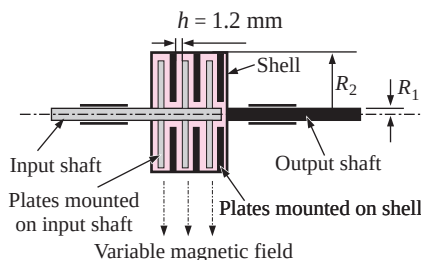


FIGURE P2-49

and (b) calculate the torque for an ER clutch with $N = 11$ for $R_1 = 50$ mm, $R_2 = 200$ mm, and $\dot{n} = 2400$ rpm if the fluid is SAE 10 with $\mu = 0.1$ Pa $\cdot$ s, $\tau_y = 2.5$ kPa, and $h = 1.2$ mm.

Answer: (b) 2060 N $\cdot$ m

2-50 The viscosity of some fluids, called magnetorheological (MR) fluids, changes when a magnetic field is applied. Such fluids involve micron-sized magnetizable particles suspended in an appropriate carrier liquid, and are suitable for use in controllable hydraulic clutches. See Fig. P2-49. The MR fluids can have much higher viscosities than the ER fluids, and they often exhibit shear-thinning behavior in which the viscosity of the fluid decreases as the applied shear force increases. This behavior is also known as pseudoplastic behavior, and can be successfully represented by Herschel–Bulkley constitutive model expressed as $\tau = \tau_y + K(du/dy)^m$. Here τ is the shear stress applied, τ_y is the yield stress, K is the consistency index, and m is the power index. For a Herschel–Bulkley fluid with $\tau_y = 900$ Pa, $K = 58$ Pa $\cdot$ s^{*m*}, and $m = 0.82$, (a) find a relationship for the torque transmitted by an MR clutch for N plates attached to the input shaft when the input shaft is rotating at an angular speed of ω while the output shaft is stationary and (b) calculate the torque transmitted by such a clutch with $N = 11$ plates for $R_1 = 50$ mm, $R_2 = 200$ mm, $\dot{n} = 2400$ rpm, and $h = 1.2$ mm.

2-51 The viscosity of a fluid is to be measured by a viscometer constructed of two 75-cm-long concentric cylinders. The outer diameter of the inner cylinder is 15 cm, and the gap between the two cylinders is 0.12 cm. The inner cylinder is rotated at 200 rpm, and the torque is measured to be 0.8 N $\cdot$ m. Determine the viscosity of the fluid.

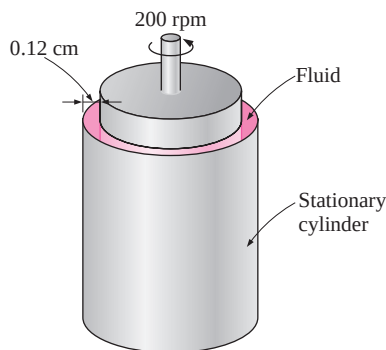


FIGURE P2-51

2-52E The viscosity of a fluid is to be measured by a viscometer constructed of two 3-ft-long concentric cylinders. The inner diameter of the outer cylinder is 6 in, and the gap between the two cylinders is 0.05 in. The outer cylinder is rotated at 250 rpm, and the torque is measured to be 1.2 lbf $\cdot$ ft. Determine the viscosity of the fluid. **Answer:** 0.000648 lb $\cdot$ s/ft²

2-53 In regions far from the entrance, fluid flow through a circular pipe is one-dimensional, and the velocity profile for

laminar flow is given by $u(r) = u_{\max}(1 - r^2/R^2)$, where R is the radius of the pipe, r is the radial distance from the center of the pipe, and $u_{\max}$ is the maximum flow velocity, which occurs at the center. Obtain (a) a relation for the drag force applied by the fluid on a section of the pipe of length L and (b) the value of the drag force for water flow at 20°C with $R = 0.08$ m, $L = 15$ m, $u_{\max} = 3$ m/s, and $\mu = 0.0010$ kg/m $\cdot$ s.

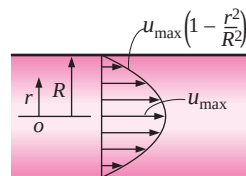


FIGURE P2-53

2-54 Repeat Prob. 2-53 for $u_{\max} = 5$ m/s. **Answer:** (b) 0.942 N

Surface Tension and Capillary Effect

2-55C What is surface tension? What is it caused by? Why is the surface tension also called surface energy?

2-56C Consider a soap bubble. Is the pressure inside the bubble higher or lower than the pressure outside?

2-57C What is the capillary effect? What is it caused by? How is it affected by the contact angle?

2-58C A small-diameter tube is inserted into a liquid whose contact angle is 110°. Will the level of liquid in the tube rise or drop? Explain.

2-59C Is the capillary rise greater in small- or large-diameter tubes?

2-60E A 0.03-in-diameter glass tube is inserted into kerosene at 68°F. The contact angle of kerosene with a glass surface is 26°. Determine the capillary rise of kerosene in the tube. **Answer:** 0.65 in

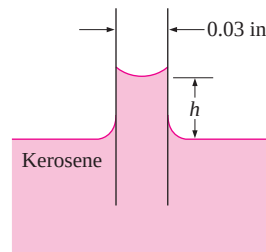


FIGURE P2-60E

2-61 A 1.9-mm-diameter tube is inserted into an unknown liquid whose density is 960 kg/m³, and it is observed that the liquid rises 5 mm in the tube, making a contact angle of 15°. Determine the surface tension of the liquid.

2-62  Determine the gage pressure inside a soap bubble of diameter (a) 0.2 cm and (b) 5 cm at 20°C.

2-63 Nutrients dissolved in water are carried to upper parts of plants by tiny tubes partly because of the capillary effect. Determine how high the water solution will rise in a tree in a 0.005-mm-diameter tube as a result of the capillary effect. Treat the solution as water at 20°C with a contact angle of 15°. *Answer: 5.75 m*

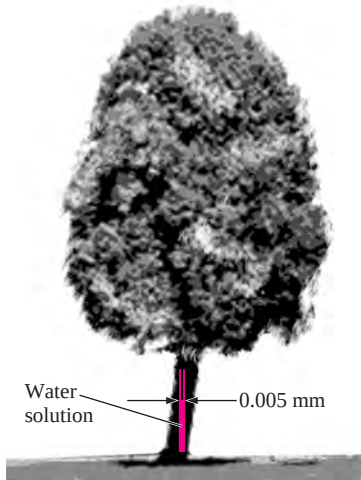


FIGURE P2-63

2-64 The surface tension of a liquid is to be measured using a liquid film suspended on a U-shaped wire frame with an 8-cm-long movable side. If the force needed to move the wire is 0.012 N, determine the surface tension of this liquid in air.

2-65 Contrary to what you might expect, a solid steel ball can float on water due to the surface tension effect. Determine the maximum diameter of a steel ball that would float on water at 20°C. What would your answer be for an aluminum ball? Take the densities of steel and aluminum balls to be 7800 kg/m³ and 2700 kg/m³, respectively.